

CDC-FM: Multi-Metric Anomaly Detection

Theoretical Foundations & Mathematical Definitions

CDC Drift AD

July 29, 2026

1. Baseline Results (VUS-PR)

Category	CDC_Raw	PCA	$d=1, k=15$
Environment	0.5487	0.5995	0.4848
Facility	0.4946	0.4366	0.5295
Finance	0.8167	0.8190	0.9097
HumanActivity	0.1549	0.0904	0.1621
Medical	0.4041	0.2034	0.4140
Sensor	0.5217	0.4517	0.5566
Synthetic	0.6356	0.6482	0.6403
Traffic	0.2880	0.3076	0.3757
WebService	0.4745	0.6981	0.6007
Overall Average	0.4779	0.5094	0.5344

Table: Average VUS-PR baseline scores across the 9 TSB-AD-U benchmark categories. The Overall Average is computed across all individual datasets. The parameter-swept baseline ($d = 1, k = 15$) provides the strongest overall competition.

2. Baseline: Raw CDC (Carré du Champ)

Displacement vector: $v = x_{nn} - x$

Transition probabilities: $P_{ij} = \frac{K_{\alpha}(x_i, x_j)}{\sum_{m=1}^k K_{\alpha}(x_i, x_m)}$

Weighted covariance ($k = 25$): $C = \sum_{j=1}^{25} P_{ij}(x_j - x_{nn})(x_j - x_{nn})^T$

Dynamic Subspace: $E_d = [e_1, \dots, e_d]$ where $\frac{\sum_{i=1}^d \lambda_i}{\sum \lambda} \geq 0.90$

Tangent projection: $v_{\parallel} = E_d E_d^T v$

Anomaly Score: $\mathcal{S}_{\text{cdc_raw}} = \|v - v_{\parallel}\|_2$

3. Baseline: Subspace CDC ($d = 1, k = 15$)

Displacement vector: $v = x_{nn} - x$

Weighted covariance ($k = 15$): $C = \sum_{j=1}^{15} P_{ij}(x_j - x_{nn})(x_j - x_{nn})^T$

Fixed Subspace ($d = 1$): $v_{\parallel} = e_1 e_1^T v$

Anomaly Score: $\mathcal{S}_{d=1,k=15} = \|v - v_{\parallel}\|_2$

4. Notation

- ▶ $x \in \mathbb{R}^d$: A sliding time-series window (test point).
- ▶ Γ_x : The local tangent plane covariance matrix at x
- ▶ $x_0 \sim \mathcal{N}(0, I)$: Gaussian latent base noise.
- ▶ $t \in [0, 1]$: Integration time (evaluated near $t = 0.95$).
- ▶ $v_\theta(x_t, t)$: The continuous normalising flow neural network predicting the restorative drift vector.

5. Standard Reconstruction (Recon_Standard)

Concept: Standard Gaussian Flow Matching without manifold constraints.

The standard interpolant:

$$x_t^{\text{std}} = tx + (1 - t)x_0$$

The predicted clean state at $t = 1$:

$$\hat{x}_1^{\text{std}} = v_\theta(x_t^{\text{std}}, t) + x_0$$

Anomaly Score:

$$\mathcal{S}_{\text{std}} = ||x - \hat{x}_1^{\text{std}}||_2$$

6. CDC Reconstruction (Recon_CDC)

Concept: Restorative Flow Matching guided by intrinsic manifold geometry.

The geometrically constrained interpolant:

$$x_t^{\text{cdc}} = tx + (1 - t)x_0 + t(\Gamma_x^{1/2}x_0)$$

The predicted clean state (removing both drift and geometric noise):

$$\hat{x}_1^{\text{cdc}} = v_\theta(x_t^{\text{cdc}}, t) + x_0 - \Gamma_x^{1/2}x_0$$

Anomaly Score:

$$\mathcal{S}_{\text{cdc}} = \|x - \hat{x}_1^{\text{cdc}}\|_2$$

7. Ensemble CDC Reconstruction (Recon_Ensemble)

Concept: Monte Carlo approximation of the CDC reconstruction error to marginalise out the stochasticity of the latent noise x_0 .

Anomaly Score:

$$\mathcal{S}_{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \|x - \hat{x}_{1,m}^{\text{cdc}}\|_2$$

where M is the ensemble size (e.g., $M = 5$), and $\hat{x}_{1,m}^{\text{cdc}}$ is the CDC prediction for the m -th independently sampled noise vector $x_{0,m} \sim \mathcal{N}(0, I)$.

8. Latent Noise Energy

Concept: Measuring the implied noise energy required to restore the point at the boundary $t \approx 1$.

Evaluated at $t = 0.99$

$$v_{\text{drift}} = v_{\theta}(x, 0.99)$$

$$z_{\text{implied}} = x - v_{\text{drift}}$$

Anomaly Score:

$$s_{\text{latent}} = \|z_{\text{implied}}\|_2$$

9. Orthogonal Energy

Concept: Measuring the raw energy of the restorative drift vector that acts orthogonally to the learned data manifold.

Evaluating the implied continuous flow drift at $t = 0.99$:

$$v_{\text{drift}} = v_{\theta}(x, 0.99)$$

Projecting v_{drift} onto the local tangent plane V :

$$v_{\parallel} = VV^T v_{\text{drift}}$$

Isolating the orthogonal movement via vector subtraction:

$$v_{\perp} = v_{\text{drift}} - v_{\parallel}$$

Anomaly Score:

$$\mathcal{S}_{\text{ortho_energy}} = \|v_{\perp}\|_2$$